

Revealing the Wavelength Dependence of Photochemical Reactions: Cutting-Edge Research in the Teaching Lab

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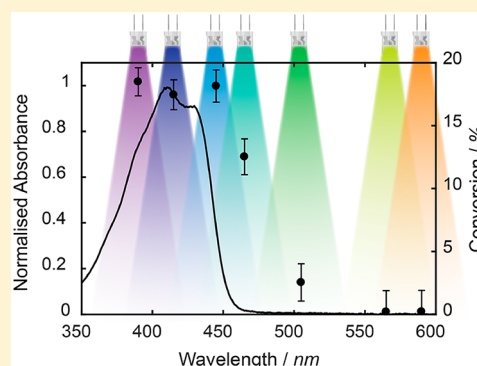
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S Supporting Information

ABSTRACT: The prompt transfer of cutting-edge science into students' curricula is a challenging task. Often research experiments are too complex or expensive to be translated into a secondary or tertiary education setting. Herein, we introduce a laboratory experiment that translates the recently developed research technique of a wavelength dependent, photochemical action plot into a student accessible format. The practical incorporates aspects of photophysics, required to calculate a constant photon emission from each different colored LED, as well as polymer chemistry in understanding the mechanism and quantitatively determining the polymerization conversion. During this practical the students (i) learn about areas where photochemical reactions are used to generate everyday materials, (ii) understand and apply the concepts of absorption and Beer–Lambert's law, (iii) record and evaluate a photochemical action plot, and (iv) critically discuss the disparate nature of action and absorption spectra to generate an understanding of the implications for photochemical material design.

KEYWORDS: Second-Year Undergraduate, Laboratory Instruction, Polymer Chemistry, Interdisciplinary/Multidisciplinary, Hands-On Learning/Manipulatives, Photochemistry, Polymerization



INTRODUCTION

Light-fueled chemical reactions have revolutionized the way everyday goods are manufactured. Examples span from the fabrication of microchips,¹ adhesives,² coatings,^{3,4} 3D printing,⁵ to dental fillings.⁶ Thus, it is important to include laboratory experimentation that explores the fundamentals of photochemical materials design into high school and undergraduate university teaching. The majority of light-based additive manufacturing processes utilize photopolymerization reactions.⁷ In these reactions, a photoinitiator absorbs light and converts the energy into a suitable chemical energy. The resulting reactive species subsequently initiate a polymerization reaction. Different colors of light (i.e., wavelengths) have different energies, and thus, not all wavelengths deliver sufficient energy to initiate a reaction.

While one might expect that the absorption spectrum of a molecule also guides the photoreactivity, our group discovered in 2017 that photoreactivity does not correlate with the absorption on a linear scale; i.e., higher extinction coefficients do not imply higher reaction yields.⁸ For example, following the initial investigation on photoinitiated polymerization

reactions, it was found that, in the case of cycloaddition reactions, the absorption maximum does not necessarily correspond to the highest photoreactivity.^{9–11} To study the wavelength dependence of photoreactivity precisely, photo-reactive molecules have to be irradiated with monochromatic light at different wavelengths using the exact same number of photons. The resulting plot, known as a “photochemical action plot”, relates the conversion of the reactive species to the irradiation wavelength, revealing critical properties for real world applications of photochemistry. These properties are important on two levels: in medical applications harmful UV light^{12,13} can be replaced by benign visible light, if the molecule is still highly reactive in the visible light regime despite low absorption, and on the other hand, manufacturers have to protect chemicals from ambient light to avoid unwanted initiation during storage or during photochemical processes.

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Access to the experimental apparatus required to record an action plot is unfortunately strongly hindered by the complexity of the photophysical setup and expensive tunable lasers, which are key to producing monochromatic light over a large continuum of the electromagnetic spectrum. To transfer the underlying principle into a university undergraduate or high school curriculum, we have developed a simple and inexpensive laboratory experiment that allows recording action plots using commercially available LEDs and basic laboratory equipment. We note that the recent paradigm shift of interrogating photoreactivity by measuring an action plot, instead of using only UV–vis spectra, as a guide for reaction design was only introduced into the literature three years ago.⁸ Thus, by translating the concept of “photochemical action plot” into a classroom environment, we provide students with the opportunity to experience and replicate cutting-edge research.

Many excellent laboratory experiments have been reported which explore the mechanisms and kinetics of various different polymerization techniques including radical polymerization,¹⁴ ATRP,^{15,16} RAFT,¹⁷ and NMP.¹⁸ Also, while there have been interesting experiments which introduce photochemistry into the undergraduate laboratory,^{19–21} there are only two recent examples which explore photoinduced polymerization.^{22,23} Of these, the experiment reported by Croisant et al.²² has a brief consideration of wavelength dependence of the photoinitiation, considering three data points only, and found that students struggled to link absorbance to its role in the mechanism of polymerization. We believe that the experiment presented herein may help to expand on this relationship by involving a more thorough wavelength analysis.

LEARNING OBJECTIVES

The experiment is designed to visualize the interplay between light and chemistry, which is the very foundation of photochemistry and today’s light-based soft matter materials design. By understanding optical concepts including absorption, Beer–Lambert’s law, and the physical limitation of broadband emission from commercially available light sources, the students learn that radical polymerizations are initiated with different efficiencies depending on the applied wavelength (color) of light.

Our educational design enables students to improve cognitive skills (i.e., application, analysis, synthesis, knowledge, and comprehension) as described in Bloom’s taxonomy.²⁴ In doing so, the students learn new techniques to improve their practical skills, such as handling laboratory equipment, but are also trained in communication and teamwork within their peer group. In addition, as the herein introduced laboratory experiment rests on recently introduced scientific procedures, it serves to demonstrate how the scientific method is applied, for example, by verifying theoretical considerations with experimental data and vice versa. The students gain insights into different research areas, specifically how chemistry and physics interplay. On the physical side, the students will learn to understand the connection among wavelength, photon number, and energy. Also, on the chemistry side, they will gain valuable experience in sample preparation, gravimetric techniques, and determination of polymerization conversion.

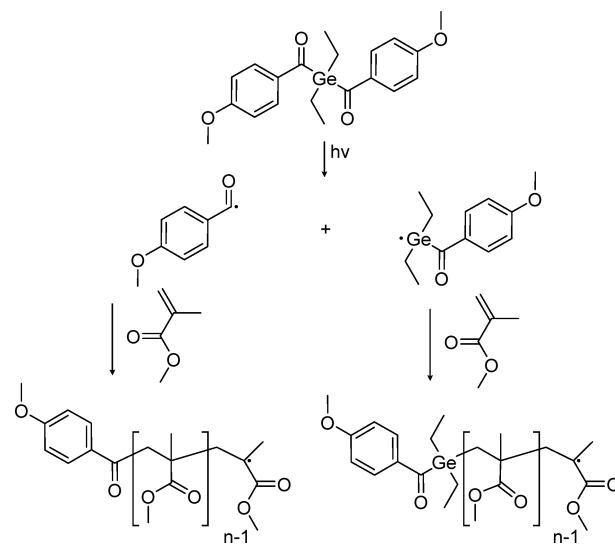
THEORY

The wavelength regions where a molecule absorbs the incoming radiation are indicated by its absorbance spectra. The amount of light absorbed can be described by Beer–Lambert’s law

$$\epsilon cl = \log_{10} \left[\frac{I_0}{I} \right] \quad (1)$$

Here, ϵ is the molar absorptivity of the molecule, c is the concentration in mol L^{−1}, l is the path length of the sample, and $\log_{10} \left[\frac{I_0}{I} \right]$ is the absorbance. One could expect that, in the wavelength regions where the sample absorbs the most light, the maximum photoreactivity would occur and, as a corollary in our case, the maximum generation of radicals and subsequent polymerization. Previous examples in literature have, however, shown that absorptivity and photochemical reactivity are not necessarily congruent.^{8–11} In the following undergraduate experiment, the students will explore the relationship between absorptivity and photochemical reactivity using the photoinitiator Ivocerin, commonly used in dental applications^{26–28} to polymerize methyl methacrylate (MMA) (refer to Scheme 1).

Scheme 1. Ivocerin Photoinitiated Polymerization of MMA



When performing action plot measurements, the use of a monochromatic light source covering the entire reactive region of the initiator, such as a tunable laser, is ideal. However, these are not widely available or appropriate for teaching applications. In recent years, LEDs across a wide range of wavelengths have been developed and can be purchased for a reasonable price. While they are not monochromatic, they are nevertheless well-suited for the purposes of the current experiment. The spectra of several commercially available LEDs used in the current experiment are shown in Figure 1, relative to the extinction spectrum of Ivocerin.

In order to accurately measure an action plot, equivalent numbers of photons must be delivered at each wavelength. Photons of different wavelengths have different energies according to Planck’s relation:

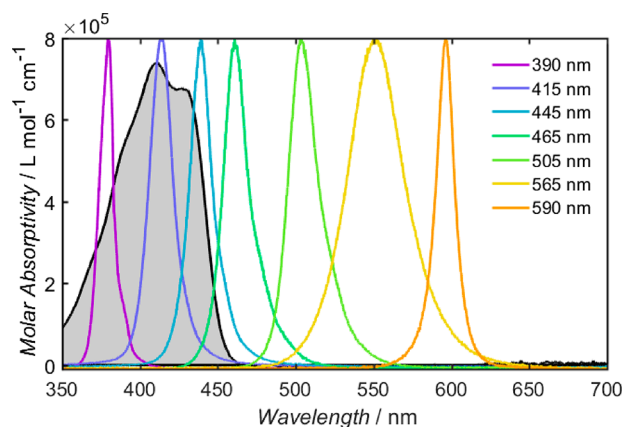


Figure 1. Spectra of available LEDs relative to extinction spectrum of Ivocerin.

$$E_p = \frac{hc}{\lambda} \quad (2)$$

Here, h is Planck's constant, c is the speed of light, and λ is the wavelength. Thus, to deliver an identical number of photons at each wavelength, the energies at each wavelength must be adjusted. The total energy delivered over a certain irradiation time can be related to the photon energy by

$$E_{\text{total}} = N_p E_p = \frac{N_p hc}{\lambda} \quad (3)$$

where N_p is the number of photons. For each measurement, E_{total} will change, but N_p must remain constant for accurate results. Practically, in the laboratory the photon number is not measured directly, but rather the power emitted, which can be related to the energy by $E_{\text{total}} = Pt$ where P is the power emitted from the LED and t is the irradiation time. The power can then be related to photon number by

$$N_p = \frac{Pt\lambda}{hc} \quad (4)$$

To maintain a specific photon number between all measurements, eq 4 can be rearranged to give the LED emission power required at each wavelength

$$P = \frac{N_p hc}{\lambda t} \quad (5)$$

To this point it is assumed that all wavelengths have similar transmission properties through the reaction vessel. Should this not be the case, eq 5 would have to include an additional constant to correct for the wavelength dependent glass transmission ($T\lambda$)

$$P = \frac{N_p hc}{\lambda t T_\lambda} \quad (6)$$

In addition, to eliminate any potential differences due to diffusion and polymerization kinetics, it is best to maintain a consistent irradiation time for all experiments.

EXPERIMENTAL OVERVIEW

The experiments were carried out in two individual sessions as part of a third year physical chemistry unit ("CVB302–Applied Physical Chemistry"), which covers topics such as electrochemistry, surface science, and polymer chemistry. As the

samples need to be dried overnight to evaporate unreacted monomer prior to gravimetry, the experiment took place over two consecutive days. Two hours were allocated on the first day for sample preparation and carrying out the experiments, while the evaluation of the data was carried out within 1 h on a following day. The students conducted the experiments in groups of two, to allow for a hands-on experience in all parts of the experiment. A detailed description of the experimental procedure can be found in the Student Manual in the [Supporting Information](#).

Sample Preparations

A stock solution of MMA and Ivocerin photoinitiator is prepared by the students. Each group of students prepares a stock solution containing a 1/50 mixture of Ivocerin and MMA. To prepare the required total of 8 samples, 8 empty vials are taken and weighed individually. The stock solution is added to each vial by weight, and the vials are weighed again. The solutions are successively purged with nitrogen from a balloon to remove any oxygen²² which may prematurely terminate the polymerization. Samples have to be kept in the dark until measurement. This was achieved by wrapping them in aluminum foil.

Calculation of Required Energies and Photochemical Reaction Setup

When measuring action plots, maintaining consistent experimental conditions across all measurements is essential. Thus, a holder, designed to perfectly fit the reaction vessel, was 3D printed. This holder is clamped centrally above a heat sink, which has the LED fixed directly (see [Figure 2a](#)).

The students started with the least efficient LED and noted the maximum output power emitted by the LED, without

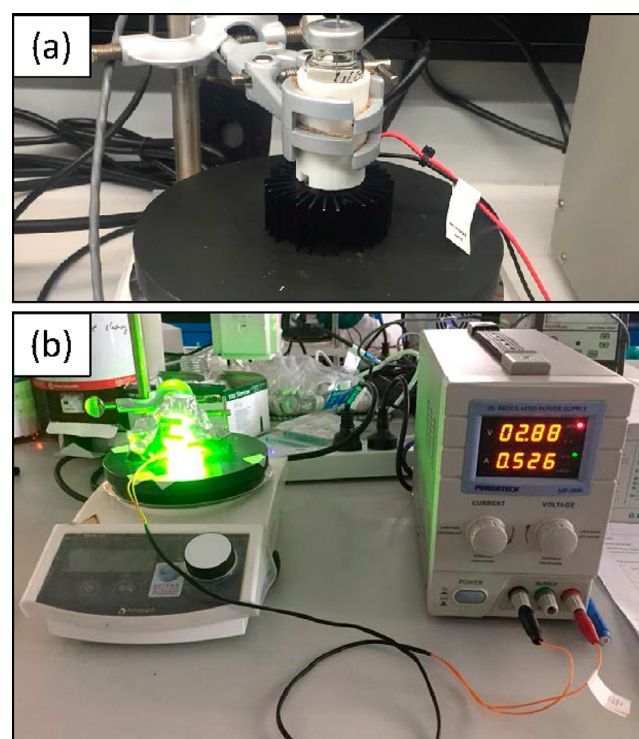


Figure 2. (a) Experimental apparatus for LED action plot measurements. LED placed centrally on stir plate with 3D printed holder clamped directly on top of LED. (b) Experimental apparatus during irradiation with 505 nm LED.

exceeding the manufacturer's specifications, using a commercially available power meter (Coherent Field Max with thermopile sensor). By inserting the power value into eq 4, the number of photons delivered over the predefined irradiation time could be determined. The students subsequently use eq 5 and the known photon number to determine the power required at each of the other wavelengths to be tested.

The students then sequentially went through the samples, irradiating each sample with a different LED as per Figure 2b, ensuring the voltage supplied to the LED was adjusted so the LED emitted the previously determined, required power. All samples were irradiated for the same amount of time, and after the irradiation the lid is immediately pierced to allow oxygen into the system and terminate the polymerization. After all samples have been irradiated, they are placed in a drying oven under reduced pressure until constant weight is achieved.

After drying overnight, all unreacted monomer is removed from the sample, allowing the students to determine the conversion of the monomer gravimetrically. After removal of the sealing lid, the samples are weighed again. On the basis of the amount of monomer remaining in the vial, the polymerization conversion can be determined using the following formula

$$C(\%) = \frac{W_{\text{post}} - W_{\text{pre}} - W_{\text{Ivocerin}}}{W_{\text{solution}} - W_{\text{Ivocerin}}} \times 100 \quad (7)$$

Here, W_{post} is the postexperiment weight, W_{pre} is the empty vial weight, W_{solution} is the amount of batch solution added, and W_{Ivocerin} is the weight of Ivocerin in the batch solution added.

The percentage of conversion for each LED wavelength tested is then plotted on a graph along with the absorbance spectra of Ivocerin. From this, the wavelength dependent polymerization is revealed and compared with the absorbance spectra.

HAZARDS

Unlike inorganic germanium complexes, organic germanium complexes have a very low toxicity.²⁹ Ivocerin is a commercially available organic germanium compound that is commonly applied in dental applications. MMA is a flammable liquid and may cause skin irritation. Nitrile gloves must be worn in addition to safety goggles. Sample preparation is to be carried out in a well-ventilated area, preferably a fume hood. Direct exposure to the UV LED should be avoided. Students are advised not to look directly into the LEDs or indirectly where light may have been scattered from other sources. While the sample holder should prevent direct contact with the bright LED light, the top of the holder should additionally be covered with aluminum foil. Students should be cautioned about the use of sharp needles.

RESULTS AND DISCUSSION

All students successfully completed the experiment, and a typical example of the evaporated samples and the action plots produced by the students are presented in Figure 3. When the reaction vessels were placed in a row, a clear variation in sample color was observed by all groups, indicating the degree of photoinitiation from the various LEDs. This was also confirmed quantitatively, from the polymer conversions determined from gravimetry, and presented in the action plot. All groups observed that the photopolymerization was

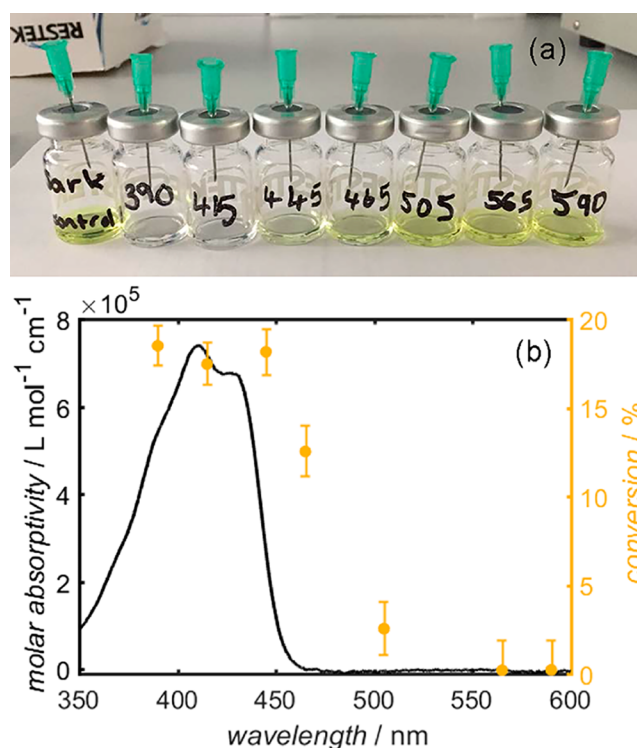


Figure 3. (a) Observed color distribution of samples after irradiation with labeled LED and (b) gravimetrically determined monomer conversions for each LED wavelength relative to the extinction spectra of Ivocerin.

efficiently initiated close to the absorption maximum of the initiator, yet, critically, also on the red side of the absorption spectrum (465 and 505 nm). The yellow and red LEDs (565 and 590 nm) produced no photoinitiation, and hence, no polymerization was observed.

To assess the applicability of the experiments in an undergraduate teaching laboratory, half of the students used a fine balance (four decimal digits) and half used a coarse balance (two decimal digits). Both achieved similar action plots, demonstrating the versatility of this experiment. Using the amounts stated in the experiment manual provided to the students, the results of the fine balance were, however, slightly more reliable. In order to use more inexpensive balances, it is recommended to increase the amount of monomer and photoinitiator by at least a factor of 5, along with the irradiation time, to guarantee accurate results.

LEARNING EXPERIENCE

For a methodical classification of learning and teaching goals, Bloom's taxonomy²⁴ has often been used. Bloom created a definition of six educational levels of activities in a cognitive domain (knowledge, comprehension, application, analysis, synthesis, and evaluation). Later on, a paradigm shift from teaching to learning orientation led to an expansion of Bloom's taxonomy to two dimensions by Anderson and Krathwohl.²⁵ The newly introduced second domain is the knowledge domain (factual, conceptual, procedural, and metacognitive knowledge). Altogether, the modern version of the expanded taxonomy combines verbs (cognitive processes) and nouns (knowledge domain), which can be arranged as per Table 1.

The various components of the practical, which focused on different cognitive and knowledge learning processes, are

Table 1. Expanded, Two-Dimensional Taxonomy^a

Knowledge Domain	Remember	Understand	Apply	Analyze	Evaluate	Create
Factual knowledge	R	E, Q1	E, Q2	Q1	Q1	Q1
Conceptual knowledge	R	E, Q1	E		Q2	Q2
Procedural knowledge	R	E	E		Q3	Q2, Q3
Metacognitive knowledge						Q2

^aFor further details, refer to the main text.

allocated within the two-dimensional revision of Bloom's taxonomy, according to Table 1. Starting with reading (R) the student's manual, the students recapitulate the basics and refresh their knowledge regarding theory. Performing the experiment (E) bridges the gap from understanding to application of factual, conceptual and procedural knowledge.

The following questions (Q1–Q3) in the postexperiment discussion were designed to support students' learning processes. Referring to the sample vials depicted in Figure 3a and the corresponding action plot depicted in Figure 3b, question one (Q1) asked the students to "Describe the different colors of the samples after irradiation. Explain why the different samples display different conversions of the monomer." In response, all students were able to relate the color change of the samples to the initiation of the polymerization and the conversion of monomer to polymer. Some, furthermore, related the extent of the color change to the degree of polymerization.

The second question (Q2) encouraged the students to think critically about the results of the action plot, asking "If your action plot does not match the extinction spectra of Ivocerin, elaborate why you think this might be?" While all students were able to link the different degrees of conversion to the photochemical activation, the students developed multiple strategies to explain the observed reactivity outside of the absorption maximum. Most students correctly identified that the LEDs used are not monochromatic and, thus, have emission overlaps with the absorption of the photoinitiator. One student recognized that Beer–Lambert's law would have a significant effect, limiting the light penetration of highly absorbed wavelengths to only a small portion of the sample. Several other students suggested the offset may be due to competing processes or reaction pathways. It is important to keep in mind that the reason for the shift in reactivity is not yet explained, so students are encouraged to be creative and come up with their own ideas. Possible explanations include the different reactivity of different excited states and the logarithmic nature of absorptivity. It is well-known that particular excited states are more likely to lead to radical formation than others, and therefore, different photon energies will produce more efficient radical formation. When the most efficient transition is offset from the absorbance maxima, even if the absorbance appears to be close to zero, the non-negligible number of absorbed photons can still produce a large number of radicals.

The final question (Q3) related to the students' factual understanding of Beer–Lambert's law to the practical ramifications in the experiment, asking "Explain how the Beer–Lambert's law would play a role in this experiment, how would you potentially overcome this?" The majority of students required prompting on this question but, after doing so, identified that the effects could be minimized, by processes such as "decreasing the path length" or "stirring the sample".

CONCLUSION

We have described an experiment to demonstrate the conversion of physical energy (light) into chemical energy (polymer formation) for students at an undergraduate level. This was accomplished by using commercially available LEDs and a simplified experimental setup, to replicate cutting-edge research techniques employed in researched laboratories. Although only performed with undergraduate students to date, the simplicity of this experiment makes it easily transferable to high school curricula.

The undergraduate students learned that radical polymerizations are initiated with different efficiencies depending on the employed wavelength, and that the absorbance spectrum is not necessarily a reliable indication of the reactivity. Instead, factors such as Beer–Lambert's law and the emission spectra of the LEDs shift the reactivity relative to the absorbance maxima. Furthermore, the students gain hands-on experience in multiple skills including sample preparation, analytic techniques and data analysis, which can be applied to chemistry, physics, and numerous other disciplines.

A major aim of our laboratory experiment was to spark interest in polymer chemistry at an early stage of the undergraduate curriculum; despite its enormous industrial importance, the focus of laboratory classes in polymer science is typically delegated to after the bachelor's degree.³⁰ Since the concepts involved in this experiment are predominantly based on Beer–Lambert's law, and not complex polymerization mechanisms, we submit that it has the potential to translate contemporary photochemistry to the classroom at the early stages.

We believe there is much to be gained by transferring the laboratory experiment to a more elementary level, e.g., one that can be used for education in high schools. Thus, there is potential to inspire high school students in the early stage of their education to consider science, technology, engineering, and mathematics (STEM) through fun and visually responsive experiments.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available on the ACS Publications website at DOI: 10.1021/acs.jchemed.9b00752.

Notes for instructors (PDF, DOCX)

Student manual (PDF, DOCX)

Example action plot calculation spreadsheet (XLSX)

Blueprint of sample holder for LED action plot (PDF)

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Notes

The authors declare no competing financial interest.

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